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Advanced Business Analytics - Final Project

B41 Bus Route Delay Forecasting

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STATEMENT OF INDIVIDUAL CONTRIBUTIONS

Team Work

- **Identifying bus route candidates based on disruptions**: After sourcing all the datasets needed for the project, each of us worked individually to identify potential candidates for our project. By performing exploratory analysis in delays and events datasets, we combined results and we ended up with B41 as the selected route.

Oier Garcia

- **Source and process of event data:** Researched and sourced relevant event data close to the bus line in the period of interest. Created a model to filter data based on proximity to bus stations of interest. Then, transformed data into useful features for the baseline and LSTM models.
- **Reforming data pre-processing notebook:** After dataset collection (bus delays, weather,events) was completed, we had to reform the data pre-processing notebook. That way we have common base before developing prediction models.
- **Contribution in the different models:** Provided support during the development of baseline and LSTM models.
- **Executive summary composition:** Extracted information and insights from all the notebooks we created. Then, used an informative design to display these findings to non-technical audiences.

Andrej Anisic

- **Data preparation notebook:** Built the pipeline converting raw GPS position records into 30-minute aggregated delay windows, handling midnight-crossing trips.
- **Feature engineering notebook:** Designed and computed all predictor variables: lag features, rolling windows, calendar dummies, and disruption persistence metrics. Integrated weather and event data into a single modelling-ready dataset.
- **Baseline models notebook:** Implemented and evaluated three forecasting approaches (Persistence, Historical Mean, Ridge Regression) across all 12 forecast horizons using a strict temporal train/validation/test split.
- **Qunatile regression notebook:** Built the initial uncertainty quantification pipeline on top of Ridge Regression outputs, implementing empirical residual intervals and quantile regression as a reference layer for the team to build on.
- **Executive summary:** Contributed the delay analysis, two-regime forecasting boundary finding, baseline model comparison results, and feature importance shift insight.

Thibaut Heim

- **Data exploration on the 2017 bus dataset from New York Open data to find the best road:** Compared the different bus lines taking into account event dataset to find the best possible line for the project.

- **Compilation of the different dataset for LSTM training:** Assembled delay date, weather data, event data, and performed feature engineering to get relevant time columns, taking into account cycles. Managed the missing data in the dataset to get a clean, complete dataset fit for LSTM training. Everything is available in the notebook "5. dataset$_{f}or_{m}odel_{t}raining.ipynb$".

- **Implementation of the Sequence to Sequence LSTM architecture:** Built the PyTorch architecture, trained it and performed inference to get visualization plots and csv outputs for the Quantile Regression. Everything is available in "6. LSTM$_{p}redictions.ipynb$".

- **Executive summary:** Contributed to the highlights and key finding sections with information from the LSTM implementation.

Vasileios Konstas

- **Dataset sourcing and initial preprocessing:** Identified and sourced the NYC bus route dataset used throughout the project. Conducted initial data preprocessing and exploratory analysis.

- **Quantile Regression notebook – Ridge Regression baseline:** Built the first QR notebook for the Ridge Regression baseline model, designing the full post-processing pipeline.

- **Quantile Regression notebook – LSTM adaptation:** Revisited and substantially revised the QR notebook to adapt the pipeline to LSTM predictions.

- **Executive summary:** Contributed the highlights section drawn from the Quantile Regression results, summarising key interval quality and disruption detection findings in accessible, non-technical language.